

Digital Logic Design Homework 3

1. Convert the following statement into a Boolean equation:

A smart home security system (S) will trigger an alarm if either the motion sensor (M) detects movement or the door sensor (D) is activated, unless the security override (O) is enabled.

$$S = (M + D) \cdot \bar{O}$$

2. Write a Boolean equation for the following condition:

A factory machine (F) starts if the power switch (P) is turned on and either the temperature sensor (T) detects a safe operating range or the emergency override (E) is engaged, but only if the maintenance mode (M) is not active.

$$F = P(E + T) \cdot \bar{M}$$

3. Construct a truth table for a function **F(A, B, C)** that outputs 1 when exactly two of the three inputs are 1, and 0 otherwise.
4. Create a truth table for a function **G(X, Y, Z)** that is 1 if **X is 1** or if **both Y and Z are 0**.
5. A system has three inputs: **A, B, and C**. The output **F** is 1 if **at least two** of the three inputs are 1. Construct a truth table for this function.
6. Design a truth table for a **majority voting system** with four voters (**A, B, C, D**). The output **V** should be 1 if **at least three out of four votes** are in favor.
7. Given the truth table below, find the **minterm expansion** of **F(A, B, C)**.

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

5.

A	B	C	output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

4.

X	Y	Z	output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

A	B	C	D	output
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

7. minterm expansion

$$F = a_0 m_0 + a_1 m_1 + a_2 m_2 + \dots + a_n m_n$$

$$F = m_1 + m_2 + m_4 + m_7$$

8. maxterm expansion

$$F = (a_0 + M_0)(a_1 + M_1)(a_2 + M_2) \cdots (a_n + M_n)$$

$$F = (0 + M_0)(1 + M_1)(1 + M_2)(0 + M_3)(1 + M_4)(0 + M_5)(0 + M_6)(1 + M_7)$$

8. Write the **maxterm expansion** for the function given in the truth table from **Exercise**

7.

$$F = M_0 M_3 M_4 M_5 M_6$$

9. Express the Boolean function $F(A, B, C) = \Sigma m(1, 3, 5, 6)$ using its **maxterm form**.

$$\Pi M(0, 2, 4, 7)$$

10. Convert the function $G(X, Y, Z) = \Pi M(0, 2, 4, 7)$ into its **minterm expansion**.

$$\Sigma m(1, 3, 5, 6)$$

7.

$$F = m_1 + m_2 + m_4 + m_7$$

8.

$$F = M_0 M_3 M_4 M_5 M_6$$

$$1. S = (M + D) \cdot \bar{0}$$

$$2. F = P(E + T) \cdot \bar{M}$$

3.

A	B	C	output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

4.

X	Y	Z	output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

5.

A	B	C	output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

6.

A	B	C	D	output
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1